Muni Zhou

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EMPLOYMENT

Dartmouth College, Department of Physics and Astronomy, Hanover, NH
Assistant Professor — Jan 2025 - now

Institute for Advanced Study, Princeton, NJ
Member in School of Natural Science — Sep 2022 - Dec 2024

Princeton University, Princeton, NJ
Princeton Presidential Postdoctoral Research Fellow — Sep 2022 - Aug 2023

EDUCATION

Massachusetts Institute of Technology, Cambridge, MA, USA
Ph.D, Plasma Physics — Sep 2016 - Aug 2022
Advisor: Nuno F. Loureiro
Thesis: Genesis, dynamics, and dissipation of turbulent magnetic fields

Zhejiang University, Hangzhou, China
B.Sc., Physics *(honors)* — Sep 2012 - Aug 2016
Advisor: Roscoe B. White
Thesis: Collisional dependence of Alfvén mode saturation in tokamaks

RESEARCH EXPERIENCE

Plasma Science and Fusion Center, MIT — 2016 - 2022
Research Assistant (Plasma Theory and Computation); Supervisor: Prof. Nuno Loureiro

Center for Integrated Plasma Studies, University of Colorado Boulder — Summer 2018, 2019
Research Assistant (Plasma Astrophysics); Supervisor: Prof. Dmitri Uzdensky

Princeton Plasma Physics Laboratory — Sep 2015 - Mar 2016
Undergraduate Research Assistant (Fusion Theory); Supervisor: Prof. Roscoe White

RESEARCH GRANTS AND COMPUTING ALLOCATIONS

PI, **NSF Grant** on Collaborative Research: Decay and Diffusion of Dynamo-Generated Magnetic Fields in Astrophysical Plasmas. PHY-2512037/PHY-2512036; Joint between Dartmouth College ($416,302) and Princeton University ($437,261). 2025-2028

Co-PI, **INCITE** allocation, Extreme Plasmas at the Spacetime Frontier: Kinetic effects in Black hole Accretion, 3M GPU node hours 2026-2029

Co-PI, **ACCESS Accelerator Project (NSF)**, Intermittency in the Transition Range: Probing the Helicity Barrier in Plasma Turbulence, 1.5M ACCESS Credits 2026-2027

PI, **ACCESS Accelerator Project (NSF)**, Generation and Relaxation of Cosmic Magnetic Fields in a Turbulent Collisionless Plasma, 2.5M ACCESS Credits 2024-2025

Leadership computing access, Frontier (ORNL), 20K GPU hours 2024-2025

AWARDS

Walter and Constance Burke Research Initiation Award 2025
Princeton Presidential Postdoctoral Research Fellowship 2022-2023
Future Investigators in NASA Earth and Space Science and Technology 2020-2022
Outstanding teaching assistant award in Department of Nuclear Science and Engineering, MIT 2019

Outstanding Bachelor's Thesis Award, Zhejiang University 2016

TEACHING AND MENTORING EXPERIENCE

Phys.122 Advanced Plasma Physics, Dartmouth College 2026

Phys.68 Introduction to Plasma Physics, Dartmouth College 2025, 2026

Phys.16 Electricity and Magnetism, Dartmouth undergraduate course for honor program 2025,2026

Lecturer for Caltech Relativistic Astrophysics Summer School 2025
Lectures on Turbulence and Dynamo.

Lecturer for PPPL Introduction to Plasma and Fusion course. 2024, 2026
Lectures on Magnetic reconnection.

Lecturer for the NSF/GPAP Summer School. 2023, 2026
Gave lectures on Fundamentals of plasmas, Linear theory of MHD waves, Magnetic Reconnection, Kinetic Plasma Physics.

SELECTED SERVICE

Vice chair of GPAP (Topical Group in Plasma Astrophysics) at APS-DPP 2025-2026

2026 AAPPS DPP Program Committee (Basic Plasma Section) 2026

2025 APS DPP Program Committee (Astrophysical and Space Sciences subcommittee) 2024-2025

Dartmouth College, Colloquium Committee and Graduate Admission Committee 2024-2025

Co-organizer, PCTS workshop on Particle transport in magnetized turbulence April 2024

Co-organizer, PCTS/PGI workshop on Improving Black Hole Accretion Models with Plasma Theory Feb 2023

MIT Plasma Science and Fusion Center Outreach Activities 2018 - 2022

Grant review for NSF, NASA FINESST, and ALCC.

Reviewer for journals including *JPP, PoP, Science Advances, Nature Communication, PNAS, ApJ, ApJL, MNRAS*

PUBLICATIONS

L. Wang, C. Dong, J. Stone, & **M. Zhou**, Impact of Parasitic Modes in Two-Dimensional Magnetorotational Instability at High Reynolds Numbers. Submitted to Astrophysical Journal Letters.

V. David, T. Adkin, J. Squire, & **M. Zhou**, Gyrokinetic theory of the Alfvénic solar-wind. Submitted to Journal of Plasma Physics.

P. Deka, F. Bacchini, **M. Zhou**, & C. Granier, Guide-Field-mediated Multiscale Instabilities in Relativistic Reconnection. Submitted to MNRAS.

H. Hakobyan, (...) (the `Entity` development team), **M. Zhou**, `Entity` – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. I: Curvilinear Special Relativistic Module. The Astrophysical Journal Supplement Series. **285**(11), (2026).

Z. Liu, **M. Zhou**, & N. F. Loureiro, Suppression of inverse magnetic energy transfer in collisionless marginally magnetized plasmas. The Astrophysical Journal Letter. **995**(2), L46 (2025).

J. Mehlhaff, **M. Zhou**, & V. Zhdankin, Radiative relativistic turbulence as an *in situ* pair-plasma source in blazar jets. *The Astrophysical Journal* **987**(2), 159 (2025).

Z. Liu, C. Silva, L. M. Milanese, **M. Zhou**, N. R. Mandell & N. F. Loureiro, Electron-only reconnection and inverse magnetic-energy transfer at sub-ion scales. *Physical Review Letter*, **134**, 155201 (2025) .

C. Zhang, C. Dong, (...) **M. Zhou**, (...) Anomalous Transient Enhancement of Planetary Ion Escape at Mars. *Nature Communications*, **16**(1), 3159 (2025).

M. Zhou, V. Zhdankin, M. W. Kunz, N. F. Loureiro & D. A. Uzdensky, Magnetogenesis in a collisionless plasma: from Weibel instability to turbulent dynamo. *The Astrophysical Journal* **960**(1), 12 (2024).

M. Zhou, Z. Liu & N. F. Loureiro, Spectrum of kinetic-Alfvén-wave turbulence: intermittency or tearing mediation? *Monthly Notices of the Royal Astronomical Society* **523**(4), 5468-5476 (2023).

M. Zhou, Z. Liu & N. F. Loureiro, Electron heating in kinetic-Alfvén-wave turbulence. *Proceedings of the National Academy of Sciences* **120**(23), e2220927120 (2023).

P. J. Catto & **M. Zhou**, Helicon and lower hybrid current drive comparisons in tokamak geometry. *Journal of Plasma Physics*, **89**(4), 905890405 (2023).

T. C. Li, Y.-H. Liu, Y. Qi & **M. Zhou**, Extended magnetic reconnection in kinetic plasma turbulence. *Physical Review Letter*, **131**(8), 085201 (2023) .

M. Zhou, V. Zhdankin, M. W. Kunz, N. F. Loureiro & D. A. Uzdensky, Spontaneous magnetization of collisionless plasmas through the action of a shear flow. *Proceedings of the National Academy of Sciences*, **119**(19), e2119831119 (2022).

M. Zhou, D. H. Wu, N. F. Loureiro & D. A. Uzdensky, Statistical description of coalescing magnetic islands via magnetic reconnection. *Journal of Plasma Physics*, **87**(6), 905870620 (2021).

P. Bhat, **M. Zhou**, & N. F. Loureiro. Inverse energy transfer in decaying, three dimensional, nonhelical magnetic turbulence due to magnetic reconnection. *Monthly Notices of the Royal Astronomical Society, 501(2), 3074-3087* (2021).

M. Zhou, N. F. Loureiro & D. A. Uzdensky, Multi-scale dynamics of magnetic flux tubes and inverse magnetic energy transfer. *Journal of Plasma Physics*, **86**(4), 535860401 (2020).

A. J. Creely, L. M. Milanese, E. A. Tolman, (...), **M. Zhou**, & A. E. White, Design study of a combined interferometer and polarimeter for a high-field, compact tokamak. *Physics of Plasmas*, **27**(4), 042516 (2020).

M. Zhou, P. Bhat, N. F. Loureiro & D. A. Uzdensky, Magnetic island merger as a mechanism for inverse magnetic energy transfer, *Physical Review Research* **1**, 012004(R) (2019).

R. A. Tinguely, A. Rosenthal, R. Simpson, (...), **M. Zhou**, B. N. Sorbom, Z. S. Hartwig & A. E. White, Neutron diagnostics for the physics of a high-field, compact, Q $\geq$ 1 tokamak, *Fusion Engineering and Design* **143**, 212-225 (2019).

M. Zhou & R. White, Collisional dependence of Alfvén mode saturation in tokamaks, *Plasma Physics and Controlled Fusion*, **58**(12), 125006 (2016).

SELECTED TALKS AND SEMINARS

Talk at Magnetic Reconnection workshop, Kagoshima, Japan June 2026
Inverse cascade in non-helical strong turbulence through electron-only reconnection.

Invited talk at NSF ECLIPSE meeting, MI May 2026
Genesis, Amplification, and Relaxation of cosmic magnetic fields.

Invited talk at Purdue meeting on High-energy Astrophysics, IN May 2026
Inverse cascade in non-helical MHD and kinetic turbulence.

Invited One-Week Lectureship, Theoretical Astrophysics Program (TAP), U. Arizona Mar 2026
multiple topics in plasma and astrophysics

Panel Discussion at the Simons collaboration/SCEECS meeting, NYC Feb 2026
Many faces of magnetic reconnection

Invited talk at the FLARE collaboration meeting , Princeton Feb 2026
Interplay between reconnection and turbulence in sub-ion range

Invited talk at AGU, New Orleans Dec 2025
Inverse cascade in KAW turbulence through electron-only reconnection

Invited talk at AAPPS Division of Plasma Physics, Fukuoka, Japan Sep 2025
Magnetogenesis in collisionless plasmas

Colloquium at University of New Hampshire Sep 2025
Magnetogenesis in collisionless plasmas

Seminar at KITP workshop Relativistic Plasma Physics: From the Lab to the Cosmos Aug 2025
Review and discussion on observations, simulations, and lab experiments towards understanding extra-galactic magnetic fields.

Colloquium at Martin A. Fisher School of Physics, Brandeis University April 2025
Magnetogenesis in collisionless plasmas and plasma-genesis in ultra-strong magnetic fields

Invited talk at Austin International Conference on Fundamental Plasma Physics Mar 2025
Magnetogenesis in collisionless plasmas

Seminar at Plasma Science and Fusion Center, MIT, MA Dec 2024
Magnetogenesis in collisionless plasmas and plasma-genesis in ultra-strong magnetic fields

Invited talk at APS Division of Plasma Physics Annual Meeting Oct 2024
Magnetogenesis in a collisionless plasma

Invited talk at High-Energy Astrophysics topical workshop at MIAPbP, Germany Sep 2024
Magnetogenesis in a collisionless plasma

Invited talk at Galaxy Cluster & Radio Relics workshop at Harvard CfA, MA Sep 2024
Magnetogenesis in a collisionless plasma

Seminar at KITP, UC-Santa Barbara, CA July 2024
Phase-space dynamics of kinetic-Alfvén-wave turbulence

Talk at Midwest Magnetic Fields Workshop at UC-Wisconsin, WI *June 2024*
Magnetogenesis in a collisionless plasma

Invited talk at Purdue Workshop on Relativistc Plasma Astrophysics, IN May 2024
Numerical study of plasma loading in the radiation-rich environment of magnetar magnetospheres

Colloquium at Dartmouth College, NH Feb 2024
How does the Universe magnetize?

Seminar at the Center for Space Physics, Boston University, MA Jan 2024
Generation and amplification of cosmic magnetic fields in a collisionless plasma

Invited talk at the Texas Symposium on Relativistic Astrophysics, Shanghai, China Dec 2023
Magnetogenesis in a collisionless plasma: from Weibel instability to turbulent dynamo

Invited talk at Modeling Plasmas Around Black Holes workshop, Leiden, Netherlands Sep 2023
Electron heating in kinetic turbulence

Plasma Astrophysics Seminar at Princeton University, NJ July 2023
Generation and amplification of turbulent magnetic fields

Tutorial seminar at the Aspen Center for Physics, CO June 2023
Turbulence and magnetic reconnection in MHD

Invited talk at the joint PGI/PCTS workshop on Improving Black Hole Accretion Models with Plasma Theory, Princeton Feb 2023
Introduction of plasma physics at black-hole accretion disks

Invited talk at CCA-Princeton Heliophysics workshop, Flatiron Institute, NY Jan 2023
Intermittency and electron heating in kinetic-Alfvén-wave turbulence

Invited talk at American Geophysical Union, Chicago, IL Dec 2022
Intermittency and electron heating in kinetic-Alfvén-wave turbulence

Heliophysics Seminar at Princeton Plasma Physics Lab Dec 2022
Intermittency and electron heating in kinetic-Alfvén-wave turbulence

Oral and poster presentation at the Solar Heliospheric and Interplanetary Environment (SHINE) meeting, HI July 2022
Intermittency and electron heating in kinetic-Alfvén-wave turbulence

Invited talk at the Magnetic Reconnection workshop, Monterey, CA May 2022
Role of magnetic reconnection in kinetic-Alfvén-wave turbulence

Invited talk at European Geophysical Union, Vienna, Austria & virtual *May 2022*
Spectrum of Kinetic-Alfvén-wave turbulence: intermittency or tearing mediation?

JPP Frontiers of Plasma Physics Colloquium July 2021
From the Weibel instability to flux-tube coalescence – a pathway to seeding plasma dynamos

Seminar at Nordic Institute for Theoretical Physics May 2021
From the Weibel instability to flux-tube coalescence – a pathway to seeding plasma dynamos

Invited talk at NASA's SolFER Spring 2021 Meeting on Solar Flare Energy Release May 2021
Statistical description of coalescing magnetic islands via magnetic reconnection

Astrophysics seminar at University of California, San Diego May 2021
From the Weibel instability to flux-tube coalescence – a pathway to seeding plasma dynamos

Astrophysics seminar at Flatiron Institute, NY Mar 2021
From Weibel instability to fluctuation dynamo in collisionless plasma simulations

Plasma physics seminar at University of Maryland, MA Jan 2021
Weibel seeding of kinetic small-scale dynamo

Invited talk at APS Division of Plasma Physics Annual Meeting Oct 2020
Multi-scale dynamics of magnetic flux tubes and inverse magnetic energy transfer via reconnection

Seminar at Institute for Fusion Theory and Simulation, Zhejiang University, China Nov 2019
Magnetic field generation and amplification in collisionless plasmas

Invited talk at AAPPS Division of Plasma Physics Annual Meeting, Hefei, China Oct 2019
Inverse magnetic energy transfer through magnetic reconnection

Inverse magnetic energy transfer through magnetic reconnection Oct 2019
Talk, APS DPP, Fort Lauderdale, FL

Invited talk at 12th Plasma Kinetics Working Meeting, Vienna, Austria Aug 2019
Multi-scale dynamics of magnetic flux tubes *(recipient of best student talk award)*

Seminar at Center for Integrated Plasma Studies, University of Colorado–Boulder Aug 2018
Inverse cascade of magnetic energy through reconnection in turbulent plasmas